

$$= 150 \text{ m s}^{-1}$$

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Ans: A

According to Malu's law

$$I = I_0 \cos^2 \theta$$

With the axes initially 90° from each other,

$$I = I_0 \cos^2(90 - \theta) = I_0 \sin^2(\theta)$$

The graph follows a sine square function.

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Ans: B